

RESEARCH ARTICLE

Wav2Lip Bridges Communication Gap: Automating Lip Sync and Language Translation for Indian Languages

VAISHNAVI VENKATARAGHAVAN¹, SHOBA SIVAPATHAM², (Member, IEEE),
AND ASUTOSH KAR³, (Senior Member, IEEE)

¹School of Electronics Engineering, Vellore Institute of Technology, Chennai 600127, India

²Centre for Advanced Data Science, Vellore Institute of Technology, Chennai 600127, India

³Department of Electronics and Communication Engineering, Dr. B. R. Ambedkar National Institute of Technology, Jalandhar, Punjab 144008, India

Corresponding author: Shoba Sivapatham (shoba.s@vit.ac.in)

ABSTRACT Translating spoken speech in videos from one language to another is known as audio-visual translation (AVT). This paper describes the implementation of an automated AVT and lip-synced dubbing application. It addresses the difficulty of synchronizing mouth movements with translated speech by building a web application that synthesizes the speaker's lip movements to match translated audio. Using ASR models, the speech from the source video is converted to text, translated into several languages, and then automatically synthesized into speech in the target language. A lip synchronization model, Wav2Lip, is used to alter the mouth movements in the video to correspond to the target language. We compare our work with two well-known ASR systems: Wav2vec 2.0 and Google Speech Recognition. Wav2vec 2.0 performs better with the lesser average WER% of 15.38 and is used in our final web application. The performance of the video dubbing component is discussed with the generated speech in Tamil, Telugu, Hindi, and English, and we determine that our generated videos outperform the existing ones. Our proposed AVT application is user-friendly for a wide variety of speakers, utilizing readily available TTS systems instead of training on an individual speaker's voice.

INDEX TERMS Wav2Lip, automatic speech recognition (ASR), audio-visual translation (AVT), lip synchronization, google speech recognition (GSR), Wav2vec 2.0.

I. INTRODUCTION

Audio-visual translation (AVT) is a subset of translation that focuses primarily on the language versioning and translation procedures used in audio-visual content and can be implemented during production or post-production [1]. The AVT refers to the process of transferring the verbal components of audio-visual works and products from one particular language to another. Implementations of AVT can be seen across films, theatrical performances, online lectures, and so on. Dubbing and voiceovers are some of the modes of translation that comprise AVT.

The associate editor coordinating the review of this manuscript and approving it for publication was Ghulam Muhammad¹.

English dominates the World Wide Web (WWW) as around 62.5% of the material available on the Internet is written in this language [2]. According to the 2011 census of India [3], only 10.67% of the total population spoke English, of which only 0.02% had it as their first language. The percentage of the Indian population using the Internet is approximately 43.4%, and this percentage is only expected to increase to approximately 62.8% [4]. This statistic reveals a gap in the amount of available content on the internet and the number of Indians who can access it.

This issue is also prevalent in other regions of the world. Only 1.08 billion out of the Earth's population of 7.9 billion speak English [5]. In several domains of technology, such as entertainment and content development, automatic voice-to-speech translation of instructional films can increase online

education and diversity. Visual cues are ideal for beginner readers, and untranslated videos without good lip sync may make for a frustrating viewing experience.

Interpreting speech and lip-syncing content that is available for the general audience, both offline and online, is usually done manually. Although, existing lip synchronization models have proposed similar ideas and can be used for lip-synced dubbing [6], [7], [8], they do not incorporate easily usable Automatic Speech Recognition (ASR), language translation, and Text-to-Speech (TTS) lip-syncing for a diverse set of speakers, all in one place. There is a lack of an accessible and intuitive system that can perform AVT automatically.

To address this issue, we propose a system for speech-to-speech and face-to-face translation using Wav2Lip. The original speaker's audio is interpreted as text and generated as translated speech. The visual information is translated by matching the speaker's lip movements to another language using pre-trained Convolutional Neural Networks (CNNs), resulting in an uninterrupted experience in the desired language for varied speakers and different angles of the face.

Our system translates English videos into a range of Indian languages, namely Tamil, Hindi, Bengali, and Telugu. However, to evaluate how well Wav2Lip works with our system, we have chosen three Indian languages to evaluate metrics against: Hindi, Telugu, and Tamil. We also evaluate English as a reference. In this paper, the aim is to achieve and address the following objectives:

- To build a web application for audio-visual translation and dubbing that translates the original video from English to Tamil, Hindi, Bengali, and Telugu.
- To determine the best ASR for our system by comparing WER metrics for Google Web Speech API (GSR) and wav2vec2.0.
- To compare different Wav2Lip lip-sync metrics for English, Hindi, Telugu, and Tamil.

The paper is organized into five sections. Section II elaborates on the proposed work, along with every system we have adopted or performed inferences for our application. Section III explains the overall architecture of the system and fleshes out the methodology behind it. The results inferred from the comparison of the ASR models (for English) and lip synchronization models (for English, Hindi, Telugu and Tamil) present in our compiled system using relevant metrics are discussed in section IV. It also discusses the resulting web application. Conclusions and future work are discussed in V.

A. RELATED WORK

We discuss previous implementations related to two types of models to study, compare, and adapt to our web application: speech-to-text (STT) and lip synchronization models. In this paper, we use the lip synchronization model that retains most of the speaker's gestures after lip synchronization and dubbing the input video. Furthermore, we conduct a study comparing two of the best open-source models:

TABLE 1. WER% of different ASR systems.

Speech Recognition Systems	WER% ¹
Google	20.63
IBM	38.11
Wit	36

¹Average WER% of Google, IBM and Wit's systems for three different speakers

Google's Web Speech API (GSR) and wav2vec2.0. The following three subdivisions explain the inferences and research gaps present in these implementations.

1) SPEECH-TO-TEXT (STT) MODELS

STT (Speech-to-Text) is the process of transcribing audio-containing speech into text in its respective language. ASR models use various machine learning methods to carry out the STT process. A system [9] proposed that would compare three different speech recognition modules: IBM's, Google's, and Wit's ASR systems. The WER (Word Error Rate), a common metric used to compare different speech recognition models, was used to compare these systems and was used for evaluation.

The authors tested these systems with three different speakers. Table 1 illustrates the average WER for all three speakers combined for the different ASR models studied [9].

The results conclude that Google's publicly available ASR system outperforms the other two because of its low WER and is a suitable model for achieving better results for transcribing speech. An unsupervised pre-training model was examined for ASR by understanding the representations of raw audio [10]. The authors trained their wav2vec model with large amounts of unlabeled audio and then used the representations made to improve the training of the acoustic model. They used a noisy contrastive binary classification task to pre-train a simple multi-layer CNN. Their model performed better than DeepSpeech2 [11], which at the time was the best-performing model for character-based ASR. An improved wav2vec2.0 framework [12], surpassed the prior model, wav2vec, on the same test data, with only one hour of labeled training data, which was 100 times lower than wav2vec.

However, no previous benchmarking study has conducted a comparison between Google's ASR system and wav2vec 2.0 for the English language. Section IV-A evaluated this using the WER metric to draw inferences from the two models.

2) LIP SYNCHRONIZATION DETECTION AND DUBBING MODELS

One of the earliest works related to this topic was discussed when Lewis [13] suggested a model for generating speech sounds, demonstrating that automatically deriving mouth movements from a speech soundtrack is a manageable issue. Phoneme recognition is made easy and precise by adapting linear prediction. Key-frames for computer-animated speech

are then created using the identified phonemes and mouth locations. The audio-video synchronization between mouth motion and speech is determined [14] in a video. They also suggested a two-stream CNN architecture that lets unlabeled data be used to learn how to embed both voice and mouth images together. They use the trained network to identify the lip-sync error in a video. When it came to lip reading and active speaker detection, they used the network to perform these two additional tasks. This implementation serves as a good lip-sync discriminator to evaluate the accuracy of lip-synced and dubbed videos. The research focuses on and creates a lip-synced video of former US President Barack Obama by setting him as the target speaker. They achieved this by deducing the sparse shapes of lips in each video frame from the audio using a Recurrent Neural Network (RNN), after which they matched the textures of the speaker's face to the lip shape. The downside of this method is that a trained model can only work for one speaker. Therefore, to achieve these results for another speaker, a dataset specific to that speaker is required, and the model must be trained again.

The research work [15] was able to generate videos of talking faces given a still image of the target speaker and an audio file. This work involved using an encoder-decoder CNN model to generate realistic talking faces. Their model was based on the AlexNet [16] and VGG-M models [17] and included an identity encoder, an audio encoder, and an image decoder. Their model did not need to be trained for a specific speaker and could be used for any face and audio. However, it could only perform inferences on still images and not on videos.

A novel approach [6] was introduced for talking face generation by combining numerous existing speech and language modules to create a workable voice-to-speech translation system that adopts and modifies the CNN architectures used in prior encoder-decoder CNN models [15] for speech-to-speech translation for a single speaker.

Another research work [7] looked into the challenge of lip-syncing an arbitrary talking face video to a particular speech piece. They discovered and presented the main factors influencing this, and they were able to address these factors by applying what they learned through an enhanced and potent lip-sync discriminator. To generate accurate lip-synced videos, [6] has improved and preferred model, Wav2Lip [7], over other existing models. Their prior work created a system that adopted speech-to-speech translation. Translating the synthetic facial map into a high-resolution and photo-realistic appearance, used two Temporal Convolutional Networks (TCN) for learning the cross-modal sequential mapping from audio inputs to lip movements [18]. It is constrained to synthesising dubbing for a singular anchor. A different approach for dubbing and large-scale AVT that translates videos between languages in [8]. In the audio and visual translation subsystems, millions of hours of data were used to train each of the large-scale generic synthesis models. Deformation Inpainting Network (DiNet) encodes the mouth

shapes to align with the audio and visual dubbing of the face that can be deformed from the input source images by incorporating mouth regions [19]. The Learned Perceptual Image Patch Similarity (LPIPS) reconstructs facial images of the speaker based on audio, produces realistic images with perfect lip synchronization, and reduces instability in the training environment. The limitation lies in the model complexity [20]

We will be working with videos with different faces and audios; the Wav2Lip implementation by [7] is adopted for the lip synchronization model to create the final speech-to-speech translation model.

B. RESEARCH GAPS

The implementation [15] falls short because it is constrained to generating videos of talking faces when provided with a still image and an audio. Since our approach relies on the gestures made by the speaker in the video, simply performing lip dubbing on a single frame would make them lose important gestures in the resulting video. The approach taken in [7] solves this issue by adopting the lip-sync discriminator and modifying the loss [14].

However, the problem of creating a system that translates a speaker as in [6] and either trains their text-to-speech system [8] again for a single speaker [6] or trains their pre-trained model again for 10,000 steps [8]. There exists a lack of a system that can be readily or easily usable. Our project aims to bridge that gap by conceptualizing and building an application that translates the audio provided and synchronize the original video to the target language by comparing and determining the best speech-to-text system available for public use by adopting and adding modifications (including processing input videos) [7].

II. PROPOSED WORK AND TECHNOLOGIES USED

The work proposed follows the following approach: Initially, the input video and audio are separated. The speaker's speech is transcribed as text using an STT model. The text is subsequently translated into the intended language using a translation module. We then use a TTS model to convert the translated text into speech. The face is then extracted from the video, and the audio is processed into Mel-frequency Cepstral Coefficients (MFCC) [21] heatmaps. The frames containing the face's lower half and the MFCC heatmaps are encoded using multiple CNN layers (Face Encoder and Audio Encoder), concatenated, and decoded and up-sampled using multiple CNN layers again (Face Decoder) to produce the results. A lip sync discriminator then judges these results, resulting in highly accurate outcomes. As a result, the synthesized lip and audio pairs are inpainted into the target face and thus a translated video is produced. Finally, the performance of these generated videos is studied by evaluating the lip-sync error between them for four different languages: English, Hindi, Telugu, and Tamil. The following

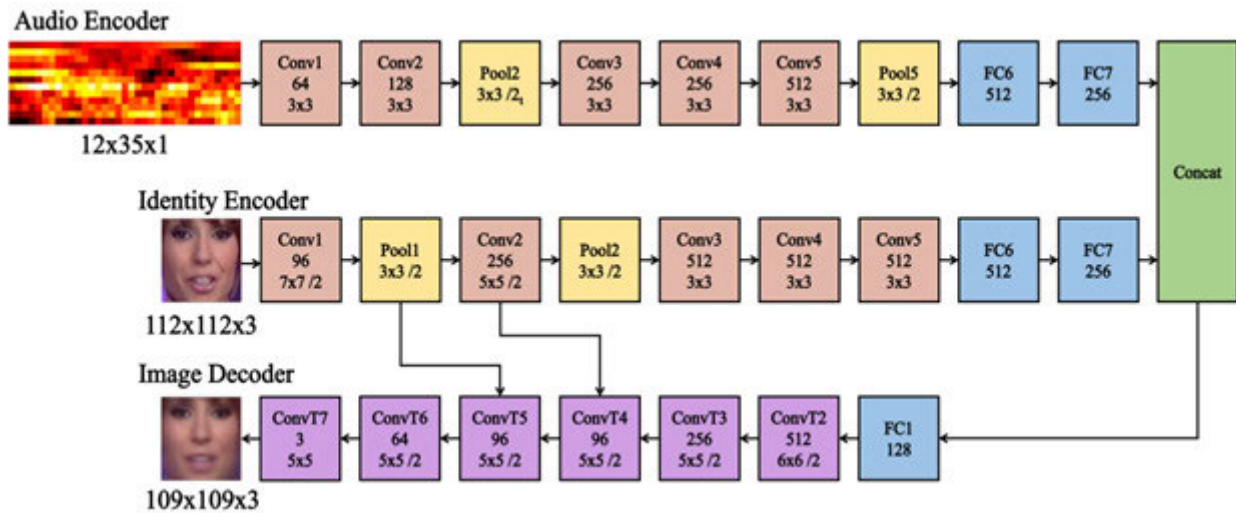


FIGURE 1. Three CNN blocks adopted from the Speech2Vid [15] model. These include, from top to bottom, the Audio Encoder, Identity (or Face) Encoder and the Image (or Face) Decoder.

sections cover some of the systems that will make up the final system.

A. AUTOMATIC SPEECH RECOGNITION (ASR) SYSTEMS

Automatic Speech Recognition (ASR) Systems work by taking the human speaker's speech signal as input and converting it into sentences as text. There are ASR systems that are phonetic based that use the Hidden Markov Model (HMM) [22] which require an N-gram language model [23] or that are "end-to-end" such as the systems that are based on Connectionist Temporal Classification (CTC) [24]. This paper uses Google's Web Speech API (GSR) [25] and wav2vec 2.0 [12] as its STT model separately.

1) GOOGLE'S WEB SPEECH API

This online API (available as Google Speech Recognition (GSR) in Python's speech recognition library) can handle both server-based and client-based/embedded recognition and synthesis. The voice recognition results are sent to the web page in the form of a list of hypotheses, each containing extra information, which we need to query to receive the required text as a result. The API in Python's speech recognition library is open source, and the ASR performs better than its counterparts [9].

2) WAV2VEC 2.0

A framework for autonomously learning representations from unprocessed audio data is provided by the wav2vec 2.0 model. This method encodes spoken sounds using a multi-layer CNN and then masks segments of the resulting latent speech approximations, akin to masked language modelling. After pre-training on unlabelled audio, the model was adjusted on labeled data with a CTC loss to be used for subsequent speech recognition tasks. It is built on the Bidirectional Encoder Representations from

Transformers (BERT) [26] architecture. The Transformer encoder is described as non-directional, and has improved upon the first wav2vec model and other ASR models [10].

B. LIP SYNCHRONISATION MODEL (WAV2LIP)

Wav2Lip [7] is a pre-trained model built on multiple CNNs. Two parts of neural networks—the lip sync discriminator (SyncNet) and the CNN audio-visual streams with a reconstruction loss—work together for processing. It also uses a discriminator to gauge image quality, to ensure the lip-sync-corrected videos produced are clear and of higher quality.

Wav2Lip uses many different residual CNN audio-visual streams and compares the generated and ground truth frames with a L1 reconstruction loss, which is used to reduce errors arising from the total of the absolute differences between actual and predicted values (seen in Equation 1). These networks include the Face Encoder, Audio Encoder, and Face Decoder, as illustrated in Figure 1. The encoders and decoders are based on earlier works [6], [15] with changes such as masking the lips of the ground-truth images to give additional pose information to the network.

$$L_1 Loss = \frac{1}{N} \sum_{i=1}^N |y_{true} - y_{predicted}| \quad (1)$$

In Equation 1, N is the batch size, y_{true} the ground-truth frame information, and $y_{predicted}$ is the generated frame information. The lower portion of the face is passed through the Face Encoder network, a CNN created with modifications to the VGG-M [17] and AlexNet [16] networks.

The video's audio is formatted into MFCC heatmaps. A MFCC heatmap describes the shape of a signal and the extracted features of an audio signal. The process of extracting these includes applying Fourier transforms (FT), mapping its powers to the mel scale (a scale of pitches),

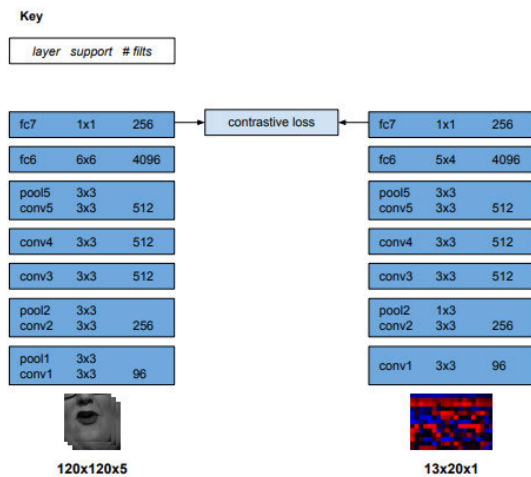


FIGURE 2. A two-stream CNN, Syncnet [14], is used to determine the quality of network frames.

taking the powers' logs, and passing it through a discrete cosine transform. These MFCC heatmaps are then passed through the audio encoder. The results of the previous two encoders are concatenated, decoded, and up-sampled using a Face Decoder [15] where they use 6 skip connections in their CNN.

Wav2Lip was originally trained with a batch size of 80 on the LRS2-BBC dataset [27]. The model was trained with the Adam Optimizer [28], using the initial learning curve of 1×10^{-4} . Using Wav2Lip, it is possible to dub videos seamlessly. It takes a video and a new audio (different from the video's audio) as inputs and gives a dubbed, lip-synchronized video with the new audio. The model is also very accessible for research, and has the highest accuracy for models trained on the LRS2 [28], the largest dataset for lip-reading sentences. Therefore, it is the best lip synchronization model we can use for our system.

C. LIP SYNC DISCRIMINATOR

SyncNet [14] (illustrated in Figure 2) is a two-stream CNN architecture that allows for the joint learning of an integration between the sound and the mouth images from unlabeled input. The trained network finds the lip-sync error in a video. The CNN for the visual stream (the video) and the audio (the speech) stream is inspired by the VGG-M network [17]. SyncNet uses the VGG-M (Early Fusion) model for this purpose because it is compact, fast to train, and has a decreased stride.

VGG-M has 3 fully connected layers and 5 convolutional layers. In the modified version used by the lip sync discriminator, the last fully connected layer and softmax layers are removed.

The loss function used in SyncNet is a contrastive loss function. The Wav2Lip model [7] uses this network to determine the quality of frames being generated by the network. Wav2Lip modifies SyncNet's architecture and uses

a cosine similarity with a binary cross entropy loss, which gives a definitive answer (0 or 1) regarding whether the generated frames are in sync or not. The paper also adds residual skip connections to their modified network.

D. TEXT-TO-SPEECH SYSTEMS (TTS)

TTS (Text-to-Speech) systems synthesize speech using the provided text. Our system has been tested with two TTS engines, namely the Google Text-to-Speech (gTTS) Module [29], which is a Python language library that allows us to use the gTTS (Google Translate's TTS) API as a voice synthesizer, and Pytttsx3 [30], which is a Python TTS conversion library that works offline. These TTS engines are readily available and provide different voices and speech speed customization (male/female or slow/fast voices).

E. TRANSLATION MODULE

The implementation adopted in this work uses the Google Translate Python module to communicate with the Google Translate Ajax API [31]. It provides for automatic language recognition as well as translations of a large bulk of text. Its capacity for mass translations and its simplicity of use for end users are the reasons for the implementation of the lip synchronization into this work.

III. ARCHITECTURE

A rough overview of the complete architecture of the system is illustrated in Figure 3. Initially, the user provides the application with a video for dubbing and translation. The input.mp4 video is processed, and the audio (English) is extracted from it using FFmpeg (Fast Forward Moving Picture Experts Group) [32]. It processes multimedia files, including the extraction of audio files from video files. The audio stream is probed and then mapped to a .wav file. The extracted audio is then fed into an ASR model, where the source language is transcribed into text. The resulting text is translated using the Google Translate Ajax API and is converted into a .wav audio format using various TTS libraries (Google's gTTS and Microsoft's SAPI5) and automatically synthesized into target language (Hindi, Tamil, etc.) speech using Wav2Lip. The system translates the visual material into the target language by synchronizing the speaker's lip movements with the translated audio. Lip synchronization processing is done by Wav2Lip, which was discussed in Section II-B. Our application translates the video into the desired language. A step-by-step overview of the system used is demonstrated in Figure 4.

A. PROCESSING THE INPUT VIDEO

The input video is processed by separating the audio from the video so that the audio can be transcribed from speech to text and translated using the Google Translate Ajax API. After this step, the audio is transcribed (using STT) to text, translated to the target language, and then synthesized to speech again (TTS), which provides us with translated audio for further processing. The application also accounts for the

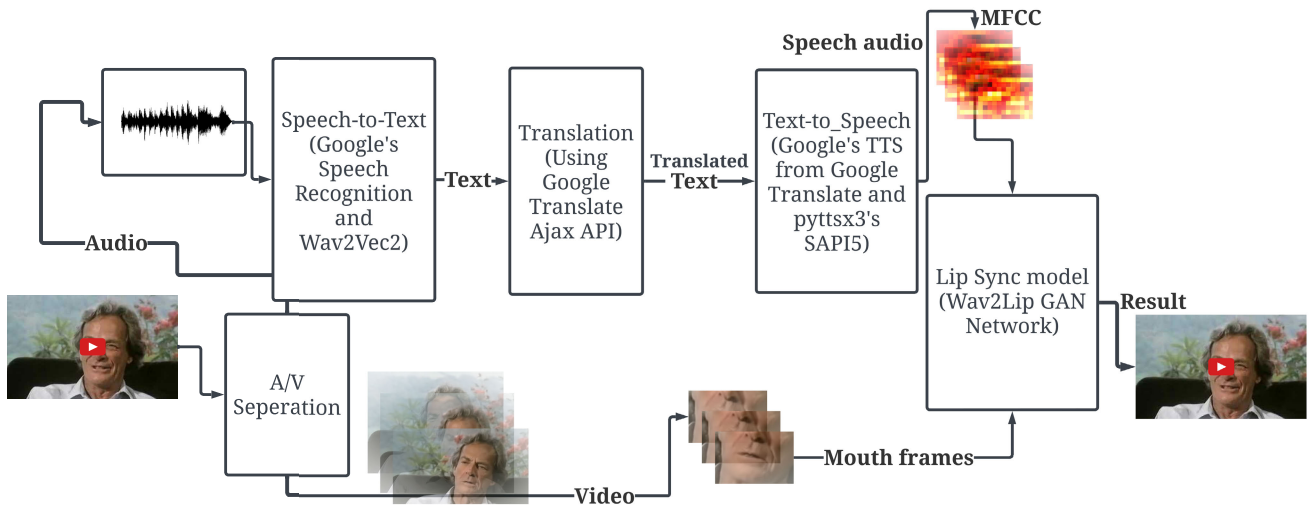


FIGURE 3. Complete architecture overview of the proposed system.

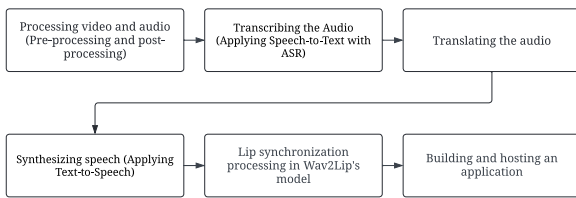


FIGURE 4. Overview of the methodology used.

mismatch between the video length and the translated audio length. This is done by decoding the different streams to get the duration, calculating the distance between the audio and video, and adjusting the frame rate or speaking rate accordingly to match the video.

Speech is slowed down or sped up by adjusting the audio tempo, and the presentation timestamp of the input frames is set accordingly using FFmpeg. The presentation timestamp is a metadata field used in MPEG (Moving Picture Experts Group) file formats that is responsible for synchronizing the speeds of the video and audio streams. The resultant video and translated audio pair are synchronized to match lip movements using lip synchronization models by running them through Wav2Lip for inference. When Wav2Lip detects no faces for synchronization, our implementation merges the translated audio and video without synchronizing any lip movements. This process is illustrated step by step in Figure 5.

B. TRANSCRIBING THE AUDIO (APPLYING STT USING ASR)

Two different ASR models are studied in this paper: the GSR API and a wav2vec 2.0 model with 960 hours of English speech training, which also functions offline. To transcribe the audio into text, the approaches taken for these models

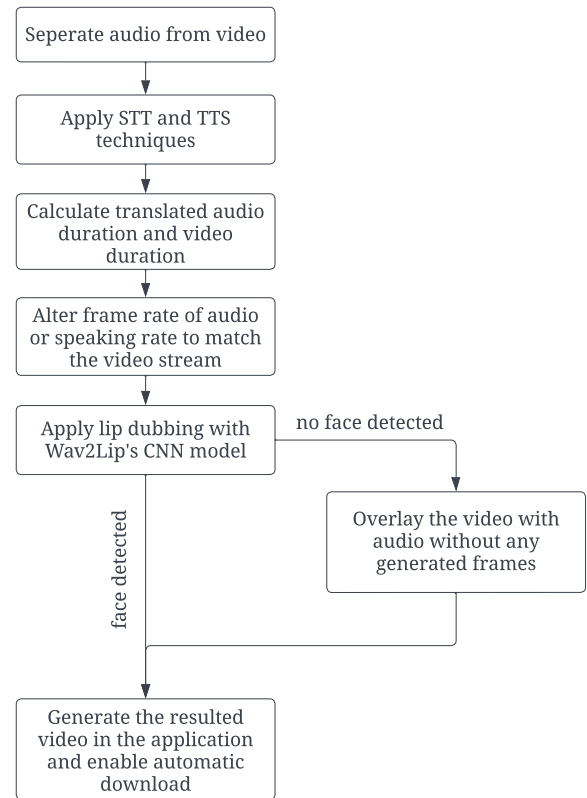


FIGURE 5. Audio and video processing performed on the target video.

differ from each other. The two approaches taken by the model are demonstrated in Figure 6.

To utilize the GSR API for transcription, the speech recognition library available in Python is adopted. The recognizer class from the speech recognition module is initialized, and the audio is recorded into the recognizer class

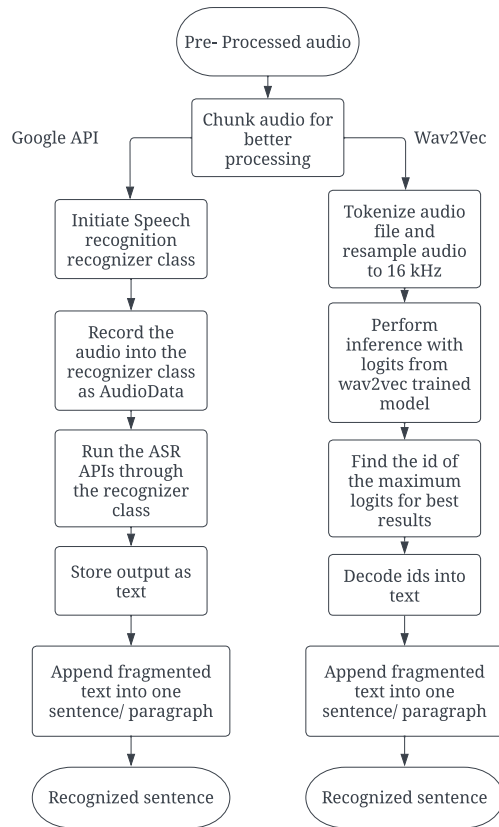


FIGURE 6. Transcribing STT with wav2vec 2.0 or GSR.

as an AudioData object. It's then provided as input to the GSR API through the recognizer class, which gives us the recognized sentence.

To utilize the trained and finetuned wav2vec 2.0 model, we tokenize the audio file and resample it to 16 kHz. Then, the audio file is used by the wav2vec 2.0 model to provide logits, which are the accuracy probabilities of the predicted texts. The logit with the highest accuracy is chosen, and the corresponding predicted text is given as the recognized sentence.

In both of these approaches, the audio can be chunked into smaller segments, and the predicted texts can be concatenated into the recognized sentence. Our system chunks the audio depending on detected silences by splitting it at any instance that is below -50 decibels. This is only required for long sentences that are processed by systems with a standard RAM of less than 12 GB to avoid an “out of memory” error.

C. SYNTHESIZING SPEECH (APPLYING TTS)

Our implementation uses two different TTS modules: gTTS and pyttsx3 (uses SAPI5 [33] voices for Windows, NSS Synthesizer for Mac OS X, and espeak [34] for other platforms). gTTS does not have the choice of having different voices for the same language, and pyttsx3's SAPI5 does not have a large range of languages. We create an application where the user can use both of these programs under

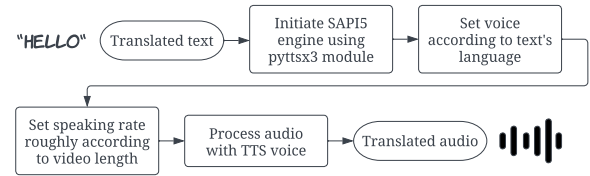


FIGURE 7. TTS Pipeline for SAPI5 voices using pyttsx3.

one system, mitigating this issue. The pipeline used for pyttsx3 is shown in Figure 7.

D. BUILDING THE WEB APPLICATION

We have developed a web application that makes speech-to-speech translation more user-friendly and accessible. Although we have only evaluated metrics for three TTS languages: English, Tamil, and Hindi in Section IV-B, we have also included support for Bengali and Telugu, as they are widely spoken languages in India. It takes four parameters and gives one video as the output. The four parameters include 3 text interfaces stating the language to be dubbed to, the preferred voice of the user (male/female), and the speed of the speaker (slow/fast), and one video selection interface where the user must upload the video file identified, and the speaker's speech is to be translated.

IV. RESULTS AND DISCUSSIONS

The ASR that works best for our model is identified, and the performance of the lip-sync model on Indian languages is evaluated in this section. The word error rate (WER) between well-known ASR models has been compared. WER is a metric that shows the distance between sentences produced by ASR models and the original sentences.

$$WER = (S + D + I) / N \quad (2)$$

Here, in Equation 2, S is the number of words that were replaced with a different one, D the number of words that were removed, I the number of words that were added, and N the total number of words in the original sentence.

Additionally, Wav2Lip's performance has been analyzed on different TTS languages (English, Hindi, and Tamil). Two new evaluation metrics in [7] and introduced by Syncnet [6], are used as metrics for this evaluation. The LSE-D metric, the Lip-Sync Error-Distance, calculates the average distance (error) between the lip and the audio given a generated video. If the distance is high (say, more than 12), that means the lip movements and audio are out of sync. The lower the distance, the better the system's performance at lip-syncing. The LSE-C metric, the Lip-Sync Error-Confidence, calculates the average confidence score of the generated video. The higher the confidence, the better the system's performance at lip-syncing. If the confidence is low (say, less than two), that means the system did not generate the best lip-synced images for the frames in the video.

The ASRs and the lip-sync model has been evaluated using a novel dataset of 120 audio-video pairs collected from various sources. The audio-video pairs are comprised of 30 input videos subjected to different audio/lip-sync methods. Figures 8, 9 and Tables 2, 3, and 4 refer to each video by the number it is assigned in the dataset. Short descriptions of each video can be found in section IV-B. With each video, 4 audio-video pairs are created with unaltered speech (the original audio), English TTS, Hindi TTS, Tamil TTS, and Telugu TTS, which results in 120 audio-video pairs, and each audio-video pair is kept in separate folders. These videos are short clips ranging from about 4 to 40 seconds long. The videos have been curated, featuring unaltered speech from a wide variety of media-sharing platforms. The dataset has 15 male and 15 female speakers for each method. It also features a diverse set of speakers in terms of ages (15-84), ethnicity, and accents (Asian, African, etc.). The videos with unaltered speech were all spoken in English by a single speaker in each video, and they have an overall duration of 7.75 minutes. The videos with altered speech were created using our application, generating 23.25 minutes of content. The total duration of the dataset is 31 minutes.

Note that this dataset does not train the existing Wav2Lip model. We use this dataset to test Wav2Lip against other languages, and note how well the model performs with videos generated by our application.

A. COMPARING WER% BETWEEN GSR AND WAV2VEC 2.0

The motivation for comparing GSR API (online) and pre-trained Wav2vec 2.0 models (offline) is that both have been shown to perform better than their counterparts, as discussed in the related works section. We benchmark these models using the WER metric.

Out of the 120 audio-video pairs present in the dataset, this inference is performed on the 30 videos with unaltered speech spoken in English. The audio are extracted from the video and is transcribed both manually and with the ASR models. The manually transcribed text is used as a reference against the automatically transcribed text to find the WER% of the generated sentences.

From the tabulated results in Table 2, it can be inferred that the wav2vec 2.0 performs better than the GSR model because it predicts more words correctly. However, wav2vec 2.0 requires extra steps to integrate since it is required to tokenize and resample input audio files to 16 kHz. On the other hand, GSR is easier to use because it's built into Python's speech recognition library. It also offers additional features such as adjusting for background noises and calibrating input sounds.

Despite these benefits, wav2vec 2.0 is consistent in performing better than GSR, beating it by a success rate of 83% (25 out of 30 videos). Therefore, we have used this wav2vec 2.0 for the resulting lip-synchronization and dubbing application.

TABLE 2. WER metrics evaluated for GSR and wav2vec 2.0.

Video	Total Words	GSR		Wav2vec 2.0	
		Incorrect Words	WER%	Incorrect Words	WER%
Video 1	41	21	51.22	3	7.32
Video 2	48	7	14.58	6	12.50
Video 3	30	12	40.00	2	6.67
Video 4	42	14	33.33	2	4.76
Video 5	74	12	16.22	10	13.51
Video 6	19	0	0.00	4	21.05
Video 7	66	16	24.24	8	12.12
Video 8	35	10	28.57	4	11.42
Video 9	46	22	47.83	4	8.70
Video 10	55	19	34.55	6	10.91
Video 11	52	12	23.08	4	7.69
Video 12	56	8	14.29	3	5.36
Video 13	44	12	27.27	5	11.36
Video 14	48	16	33.33	3	6.25
Video 15	32	8	25.00	3	9.38
Video 16	14	4	28.57	3	21.43
Video 17	23	5	21.74	4	17.39
Video 18	49	9	18.37	4	8.16
Video 19	20	5	25.00	2	10.00
Video 20	18	18	100.00	18	100.00
Video 21	39	11	28.21	5	12.82
Video 22	25	3	12.00	0	0.00
Video 23	22	5	22.73	5	22.73
Video 24	20	12	60.00	4	20.00
Video 25	24	9	37.50	2	8.33
Video 26	34	7	20.59	5	14.71
Video 27	9	0	0.00	0	0.00
Video 28	28	3	10.71	0	0.00
Video 29	34	25	73.53	11	32.35
Video 30	9	2	22.22	4	44.44
Average WER%			29.82		15.38

The best performing WER for each video has been highlighted and the average WER% is determined.

B. ANALYZING LIP-SYNC ERROR ON AUTOMATICALLY GENERATED SPEECH

To evaluate how well the Wav2Lip-based model works with our proposed system, we evaluated 84 audio-video pairs with the LSE-D and LSE-C metrics.

The metrics in Tables 3 and 4 are obtained automatically using the SyncNet model that determines the confidence and distance of the audio-video pair, using evaluation scripts to extract this metric. We generated 30 lip-synced videos, each with its original audio in English, and then with dubbed English TTS, Tamil TTS, Telugu TTS and Hindi TTS audio. SyncNet evaluates videos by recognizing 'scenes,' and some of the videos did not have enough content to be evaluated. Videos that were not evaluated for LSE-D and LSE-C include: Video 2 - Introductory speech, Video 3 - Emma Watson speech, Video 4 - Emma Watson speech 2, Video 11 - Priyanka Chopra speech, Video 16 - Steve Harvey clip, Video 22 - Monologue about distractions, Video 23 - Captain America dialogue, Video 27 - Malala Yousafzai speech, Video 30 - Interview clip. Out of the 120 lip-synced videos we made for our dataset, 90 were used for inference. SyncNet was able to recognize 118 scenes from 84 audio-video pairs, including 21 videos of unaltered English speech, English TTS, translated Hindi TTS, translated Telugu TTS,

TABLE 3. LSE-D and LSE-C metrics evaluated for TTS audio-video pairs for untranslated original reference videos with natural speech.

Videos Evaluated	LSE-D ↓	LSE-C ↑
Video 1 ¹ - Richard Feynman interview	7.552	6.763
Video 5 - Public speaking tutorial 1	6.489	9.546
Video 6 - Public speaking tutorial 2	7.193	7.212
Video 7 - Etiquette tutorial	6.472	9.244
Video 8 - TEDx Talk	7.644	6.446
Video 9 - Elder woman narration	8.685	4.613
Video 10 - Stand-up comedy	7.594	5.703
Video 12 - Greta Gerwig speech	5.703	6.347
Video 13 - United Nations speech	7.722	5.490
Video 14 - Little Women movie clip	9.223	5.628
Video 15 - Charlie Chaplin monologue	8.718	4.895
Video 17 - Mark Zuckerberg AD 1	5.859	10.370
Video 18 - Mark Zuckerberg AD 2	6.710	9.056
Video 19 - Independence Day movie clip	11.511	1.569
Video 20 - Sports channel clip	8.165	3.785
Video 21 - TED Talk	6.841	7.670
Video 24 - Indian news channel clip	8.346	5.860
Video 25 - Al Pacino monologue	11.332	1.879
Video 26 - Steve Jobs interview	7.653	6.801
Video 28 - UCLA Alumni Speech	8.108	5.882
Video 29 - Algorithms tutorial	8.248	4.778

¹ Videos in our dataset start with a number.

TABLE 4. LSE-D and LSE-C metrics evaluated for TTS audio-video pairs.

Videos Evaluated	English ¹		Hindi ²		Tamil ²		Telugu ²	
	LSE-D ↓	LSE-C ↑	LSE-D ↓	LSE-C ↑	LSE-D ↓	LSE-C ↑	LSE-D ↓	LSE-C ↑
Video 1	7.07	7.99	7.25	7.33	7.04	7.68	7.32	6.96
Video 5	6.97	8.43	7.13	7.98	7.98	6.52	7.12	6.32
Video 6	6.27	8.62	6.32	9.17	6.62	7.27	7.18	7.25
Video 7	6.59	8.51	6.46	8.57	6.81	7.79	6.32	7.00
Video 8	6.07	9.63	6.64	8.44	6.38	7.72	8.13	6.99
Video 9	5.88	10.05	6.44	8.74	6.23	8.35	7.45	7.52
Video 10	6.80	8.86	6.82	8.99	7.26	7.95	8.25	7.82
Video 12	6.45	9.62	6.47	8.98	6.99	7.58	6.75	8.10
Video 13	6.69	8.90	6.93	7.78	6.89	7.52	6.12	7.24
Video 14	8.35	7.21	8.13	6.72	8.38	5.94	6.88	6.11
Video 15	5.72	9.98	7.04	7.71	6.61	8.21	5.84	6.93
Video 17	6.85	7.75	7.21	7.30	6.86	7.10	7.15	6.43
Video 18	6.77	8.33	7.64	7.16	7.35	6.75	6.67	6.59
Video 19	11.68	3.48	11.04	3.40	11.04	3.51	8.23	4.73
Video 20	6.02	9.57	6.71	8.50	6.64	7.66	8.78	6.93
Video 21	6.46	8.35	7.73	6.72	7.49	6.55	8.58	7.92
Video 24	6.11	9.26	6.90	8.13	6.90	7.40	8.12	6.82
Video 25	6.49	8.69	12.03	1.87	12.25	1.58	9.55	4.89
Video 26	6.22	9.19	7.26	7.58	6.66	8.07	7.15	8.66
Video 28	6.86	7.58	6.68	9.07	7.02	7.46	5.84	6.77
Video 29	6.99	7.49	8.25	5.46	7.45	6.18	8.10	9.38

Lip-synced TTS videos are (21 each of English, Hindi, Telugu, and Tamil TTS) evaluated by Wav2Lip's metrics. The best performing LSE-D and LSE-C for each video variant have been highlighted.

¹ These videos have not been translated using Google Translate Ajax API.

² These videos have been translated from English using Google Translate Ajax API.

and translated Tamil TTS. The obtained results are tabulated in Table 5 along with inference taken from Wav2Lip's performance on the ReSyncEd dataset [7]. The average LSE-D and LSE-C metrics of these videos are calculated on our novel TTS dataset dubbed with different Indian languages, using our system along with the original reported metrics for the ReSyncEd TTS dataset [7].

From Table 5 results, it can be concluded that the lip-synced videos did better than the reference videos that are not lip-synced or dubbed. All four types of lip-synced

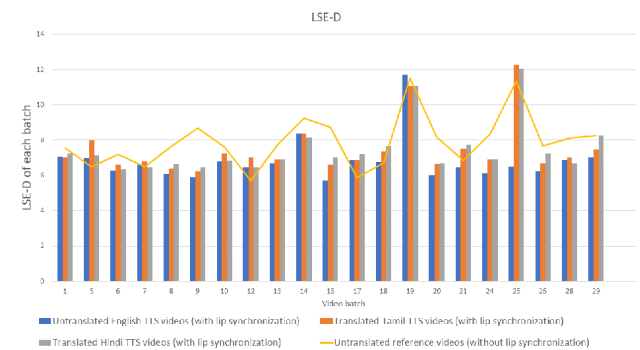
TABLE 5. Benchmarking Wav2Lip for different languages.

Video type	LSE-D ↓	LSE-C ↑	Overall Experience ³
Untranslated reference videos ¹	7.894	6.191	
Untranslated English TTS videos	6.826	8.454	3.6
Translated Hindi TTS videos	7.481	7.411	3.6
Translated Tamil TTS videos	7.470	6.894	3.3
Translated Telugu TTS videos	7.406	7.017	3.0
ReSyncEd TTS videos ²	6.659	8.126	

¹ This video type has not been lip-synced with our system and is for reference.

² Source: [7]

³ Only recorded for the lip-synced TTS videos

**FIGURE 8.** Combined Selection Interface.

TTS videos performed well, but the untranslated videos with lip-synced English TTS did better than the others, having a score similar to the metrics generated by the ResyncEd TTS dataset with an average LSE-D of 6.826 and an average LSE-C of 8.454. The untranslated reference videos in English performed the worst, with an average LSE-D of 7.894 and an average LSE-C of 6.191. As the lip-synchronization of the chosen Indian languages performed better than the untranslated reference videos without lip-synchronization or dubbing, we can infer that our system can be used to lip-sync TTS-dubbed videos in Indian languages accurately. It can be inferred from the charts with the metrics of the 21 audio-video batches (labelled by video number) shown in Figures 8 and 9 that, on average, the lip-synced videos have a lower LSE-D and a higher LSE-C than the reference videos.

Though the evaluation of our videos showed favorable results, it is still important to take human experience into account. We asked 10 evaluators (4 men and 6 women) from different backgrounds and ages (20-60 years) to rate the generated lip-sync videos on overall experience. The evaluators have given moderate ratings for the English and Hindi TTS videos at 3.6. The Tamil TTS videos was rated the lowest at 3.3. The Telugu TTS video was rated at 3.0. Evaluators unanimously agreed that dubbed TTS voices impacted the viewing experience due to the

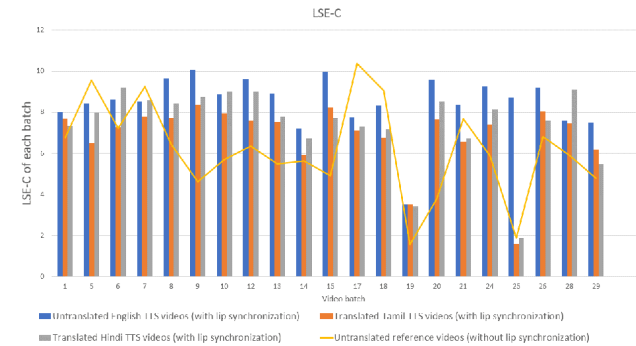


FIGURE 9. Video output produced by application.

TABLE 6. State-of-the-art approaches with our dataset.

Model Used	LSR-2 Dataset		Our Dataset	
	LSE-D	LSE-C	LSE-D	LSE-C
Wav2Lip [7]	7.151	6.761	5.412	5.817
LipGAN[6]	10.33	3.199	8.142	4.728
Wav2Lip+GAN [7]	6.469	7.781	6.314	5.984
DiNet [19]	8.377	5.377	7.338	6.028
LPIS - AttWav2Lip [20]	7.171	6.834	6.770	6.185
Proposed work	8.975	7.917	7.895	6.168

robotic tone of TTS. However, they also agreed that the lip-synced dubbed videos were easy to follow and helped them understand the content better.

C. STATE-OF-THE-ART APPROACHES

The lip synchronization was evaluated with the two metrics LSE-C and LSE-D to produce the confidence and distance scores. The proposed work has been compared with the SOTA methods, and the average results are produced for the benchmark LRS2 dataset and our (120 audio-video pairs) dataset. The proposed work outperforms existing methods and shows the stability of faces with consistent videos. The LipGAN produces a high score for our dataset for LSE-D metrics, whereas the LSE-C superperforms all the existing methods for the real-time dataset we collected. From Table 6, it is inferred that the proposed work achieves significant results for our dataset

D. WEB APPLICATION IMPLEMENTATION

The resulting application is demonstrated through two practical examples. In the first scenario, a user interacts with the web application by selecting a video featuring an English-speaking male speaker, as shown in Figure 10. The system performs real-time multilingual dubbing by translating the speech into Hindi, adjusting the speaking pace to a slower setting, and generating a male voice to match the speaker’s original gender. The real time dubbing enables the speakers to be understood across different languages instantly, while accurate lip sync ensures a natural viewing experience. The output—an automatically dubbed video where the original speaker now appears to speak Hindi in real

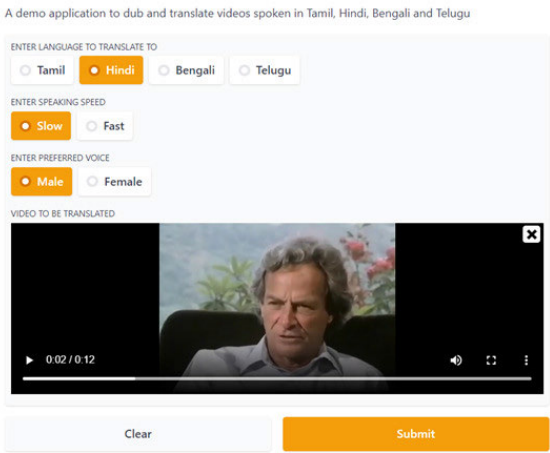


FIGURE 10. Combined Selection Interface.

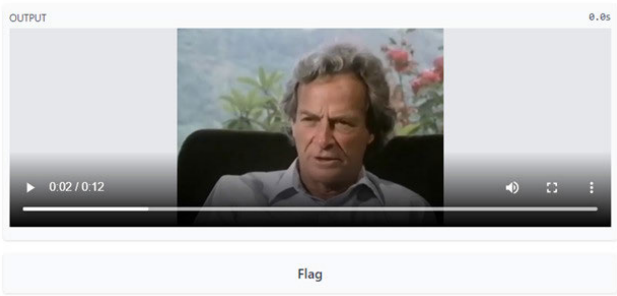


FIGURE 11. Video output produced by application.

time—is illustrated in Figure 11. This showcases the potential of our system for live international communication, such as during webinars or conferences. These examples demonstrate how AVT technologies can significantly enhance communication and inclusivity in global, time-sensitive contexts. The web application is publicly accessible via our GitHub repository.

Our application produces resulting videos that are synthesized, raising questions regarding privacy concerns. While there is scope of TTS improvement in our application, currently our system dubs its speakers with robotic TTS, which makes the overall video less realistic for potential misuse. However, we believe it is important to be wary of misuse of synthesized data and the grave implications of not following fair use when dealing with generated content.

V. CONCLUSION AND FUTURE WORK

The work describes the design approach taken, technologies, and methodology used for building an easy-to-use web application that translates videos to Indian languages in detail. Two popular STT modules were used: GSR (Web-Speech API) and wav2vec 2.0. The latter was picked for use in our application since it performed better with a WER% of 15.38. To evaluate the videos generated by our application, a novel dataset was created containing lip-synced

videos in English, Hindi, Tamil and Telugu along with their original reference videos in English. The LSE-D and LSE-C of 118 generated scenes from 84 audio-video pairs were evaluated. Since the lip-synced videos performed better than the reference videos and received a favourable response from evaluators, it can be concluded that our application generates translated videos of good quality. The application developed is easy to use and can perform speech-to-speech translation from English to four Indian languages with just a few simple inputs from the user. This includes languages that were tested for their performance such as Hindi, Telugu and Tamil, alongside other languages such as Bengali and Telugu.

There are limitations in our work that needs to be discussed as well. We use Wav2Lip's pretrained model without fine-tuning the model or training its lip-sync discriminator, keeping it as-is for our application. There are 22 official languages in India, and yet our work only performs inferences on two of them due to the lack of different readily available TTS language voices. Therefore, there is plenty of scope for further studies in this area. The lip-synchronization model could be improved by further training the lip-sync discriminator and by improving upon the multiple convolutional networks present in the Wav2Lip model. Also, thereal-time processing speed or incorporating other sequential-data-processing deep learning architectures such as LSTM-based or Transformer-based translation models can be implemented for further enhancements. Similarly, more Indian languages can be analyzed to make other inferences. Furthermore, natural-sounding TTS can be integrated to increase the audience's overall experience.

Additionally, there are many practical applications for this work as well. When implemented on a larger scale, like being integrated into media sharing platforms, it can make educational videos and urgent political videos in English more accessible to people who don't speak the language. This would facilitate the spread of knowledge faster throughout India. Moreover, English can be dubbed into other European languages such as French or Italian, or East Asian languages such as Japanese or Mandarin, or many other languages across the world.

DECLARATIONS

COMPETING INTERESTS

The authors declare that they have no competing interests.

AUTHOR'S CONTRIBUTIONS

Vaishnavi Venkataraghavan created the dataset and the application, and did the programming for the project. Shoba Sivapatham analyzed and evaluated the results of the project, and also conceived the presented idea. Vaishnavi Venkataraghavan and Prathulyan Nagalingam wrote the manuscript and conducted literature reviews. Asutosh Kar supervised the project and helped compare with existing articles.

AVAILABILITY OF DATA AND MATERIALS

The source code and the videos used for this article can be found on <https://github.com/thelonelyextrovert/avt-indic>. Wav2Lip, the model this implementation is built upon, can be found on <https://github.com/Rudrabha/Wav2Lip>. The application built for this project can be found on <https://huggingface.co/spaces/capstonedubtrack/Indiclanguageubbing>

REFERENCES

- [1] D. Chiaro, "Audiovisual translation," in *The Encyclopedia of Applied Linguistics*. Blackwell Publishing Ltd, 2012. [Online]. Available: <https://onlinelibrary.wiley.com/doi/10.1002/9781405198431.wbeal0061.pub2>
- [2] (Apr. 2022). *Usage Statistics of Content Languages for Websites*. [Online]. Available: https://w3techs.com/technologies/overview/content_language
- [3] Census India. (2011). *C-17 Population by Bilingualism and Trilingualism*. [Online]. Available: <https://web.archive.org/web/20191113211224/http://www.censusindia.gov.in/2011census/C-17.html>
- [4] H. Sheth. (Jun. 2021). *India's Active Internet Population Likely to Reach 900 Million by 2025: Report*. [Online]. Available: <https://www.thehindubusinessline.com/info-tech/indias-active-internet-population-likely-to-reach-900-million-by-2025-report/article34714569.ece>
- [5] D. Eberhard, G. F. Simons, and C. D. Fennig, *Ethnologue*. Dallas, TX, USA: SIL International Publications, 2022.
- [6] K. R. Prajwal, R. Mukhopadhyay, J. Philip, A. Jha, V. Namboodiri, and C. V. Jawahar, "Towards automatic face-to-face translation," in *Proc. 27th ACM Int. Conf. Multimedia*, Oct. 2019, pp. 1428–1436.
- [7] K. R. Prajwal, R. Mukhopadhyay, V. P. Namboodiri, and C. V. Jawahar, "A lip sync expert is all you need for speech to lip generation in the wild," in *Proc. 28th ACM Int. Conf. Multimedia*, Oct. 2020, pp. 484–492.
- [8] Y. Yang, B. Shillingford, Y. Assael, M. Wang, W. Liu, Y. Chen, Y. Zhang, E. Sezener, L. C. Cobo, M. Denil, Y. Aytar, and N. de Freitas, "Large-scale multilingual audio visual dubbing," 2020, *arXiv:2011.03530*.
- [9] F. Filippidou and L. Moussiades, "A benchmarking of IBM, Google and wit automatic speech recognition systems," in *Proc. IFIP Int. Conf. Artif. Intell. Appl. Innov.* Cham, Switzerland: Springer, 2020, pp. 73–82.
- [10] S. Schneider, A. Baevski, R. Collobert, and M. Auli, "wav2vec: Unsupervised pre-training for speech recognition," in *Proc. Interspeech*, Sep. 2019, pp. 1–12.
- [11] D. Amodei et al., "Deep speech 2: End-to-end speech recognition in English and Mandarin," in *Proc. Int. Conf. Mach. Learn.*, Jan. 2015, pp. 173–182.
- [12] A. Baevski, H. Zhou, A. Mohamed, and M. Auli, "Wav2vec 2.0: A framework for self-supervised learning of speech representations," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2020, pp. 12449–12460.
- [13] J. Lewis, "Automated lip-sync: Background and techniques," *J. Visualizat. Comput. Animation*, vol. 2, no. 4, pp. 118–122, Oct. 1991.
- [14] J. S. Chung and A. Zisserman, "Out of time: Automated lip sync in the wild," in *Proc. Asian Conf. Comput. Vis.*, Jan. 2017, pp. 251–263.
- [15] A. Jamaludin, J. S. Chung, and A. Zisserman, "You said that?: Synthesising talking faces from audio," *Int. J. Comput. Vis.*, vol. 127, nos. 11–12, pp. 1767–1779, Dec. 2019.
- [16] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, no. 6, pp. 84–90, May 2017.
- [17] K. Chatfield, K. Simonyan, A. Vedaldi, and A. Zisserman, "Return of the devil in the details: Delving deep into convolutional nets," 2014, *arXiv:1405.3531*.
- [18] R. Zheng, Z. Zhu, B. Song, and C. Ji, "A neural lip-sync framework for synthesizing photorealistic virtual news anchors," in *Proc. 25th Int. Conf. Pattern Recognit. (ICPR)*, Jan. 2021, pp. 5286–5293.
- [19] Z. Zhang, Z. Hu, W. Deng, C. Fan, T. Lv, and Y. Ding, "DINet: Deformation inpainting network for realistic face visually dubbing on high resolution video," in *Proc. AAAI Conf. Artif. Intell.*, Jun. 2023, vol. 37, no. 3, pp. 3543–3551.
- [20] Z. Chen, X. Wang, L. Xie, H. Yuan, and H. Pan, "LPIPS-AttnWav2Lip: Generic audio-driven lip synchronization for talking head generation in the wild," *Speech Commun.*, vol. 157, Feb. 2024, Art. no. 103028.

- [21] S. Davis and P. Mermelstein, "Comparison of parametric representations for monosyllabic word recognition in continuously spoken sentences," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. ASSP-28, no. 4, pp. 357–366, Aug. 1980.
- [22] L. R. Rabiner and B. Juang, "An introduction to hidden Markov models," *IEEE ASSP Mag.*, vol. 3, no. 1, pp. 4–16, Jan. 1986.
- [23] P. F. Brown, V. J. Della Pietra, P. V. deSouza, J. C. Lai, and R. L. Mercer, "Class-based n-gram models of natural language," *Comput. Linguistics*, vol. 18, no. 4, pp. 467–480, 1992.
- [24] A. Graves, S. Fernández, F. Gomez, and J. Schmidhuber, "Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks," in *Proc. 23rd Int. Conf. Mach. Learn. (ICML)*, 2006, pp. 369–376.
- [25] J. Adorf, "Web speech API," *KTH Roy. Inst. Technol.*, vol. 1, pp. 1–15, Mar. 2013.
- [26] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," 2018, *arXiv:1810.04805*.
- [27] T. Afouras, J. S. Chung, A. Senior, O. Vinyals, and A. Zisserman, "Deep audio-visual speech recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 12, pp. 8717–8727, Dec. 2022.
- [28] J. C. Duchi, E. Hazan, and Y. Singer, "Adaptive subgradient methods for online learning and stochastic optimization," *J. Mach. Learn. Res.*, vol. 12, no. 61, pp. 2121–2159, Feb. 2011.
- [29] P. N. Durette. (Mar. 2022). *Google Text-to-Speech*. [Online]. Available: <https://gtts.readthedocs.io/en/v2.2.4/>
- [30] N. M. Bhat. (Jul. 2021). *Pytsx3 Documentation*. [Online]. Available: <https://buildmedia.readthedocs.org/media/pdf/pytsx3/latest/pytsx3.pdf>
- [31] S. Han. (2017). *Free and Unlimited Google Translate API for Python*. [Online]. Available: <https://py-googletrans.readthedocs.io/en/documentation>
- [32] (2022). *FFmpeg*. [Online]. Available: <http://ffmpeg.org/>
- [33] Microsoft. (Apr. 2012). *Microsoft Speech API (SAPI) 5.3*. [Online]. Available: [https://docs.microsoft.com/en-us/previous-versions/windows/desktop/ms723627\(v=vs.85\)](https://docs.microsoft.com/en-us/previous-versions/windows/desktop/ms723627(v=vs.85))
- [34] Sourceforge. (2022). *Espeak Text to Speech*. [Online]. Available: <http://espeak.sourceforge.net/>



VAISHNAVI VENKATARAGHAVAN received the Bachelor of Engineering degree from VIT Chennai. She is currently pursuing the master's degree with the University of Michigan. She has experience in creating, automating, and maintaining dashboards in PowerBI. She was a Principal Analyst with AB InBev and completed her Internship as a Graphic Designer. Her expertise lies in data analysis and illustration, and she also dabbled in ML, AI, and creating bots/designing apps.



SHOBA SIVAPATHAM (Member, IEEE) received the Ph.D. degree from Anna University. She completed the Postdoctoral Fellowship from IIITDM Kancheepuram. She is currently an Associate Professor with the Centre for Advanced Data Science, Vellore Institute of Technology, Chennai. She has more than 42 research articles in areas of image processing, deep learning, artificial intelligence, machine learning acoustic signal processing, speech enhancement, speech separation, agriculture, heritage, medical disease prediction, and speech recognition at international levels. She is also involved in many scholarly services, such as reviewing research papers for international peer-reviewed journals and conferences. She is an Active Member of number of professional bodies, such as IEEE and the ISTE Society.



ASUTOSH KAR (Senior Member, IEEE) is an Associate Professor with the Department of ECE, Dr. B. R. Ambedkar National Institute of Technology Jalandhar, Jalandhar, India, since 2023. He was an Assistant Professor (Grade-I) with Indian Institute of Information Technology, Design and Manufacturing, Kancheepuram, Chennai, India, and also with the Birla Institute of Technology and Science, Pilani. He is also a Visiting Research Scientist with the ESAT Laboratory, Catholic University of Leuven, Belgium, King Mongkut's Institute of Technology Ladkrabang, Bangkok, Thailand, University of Kragujevac, Serbia, Kragujevac, Serbia, and University of Oldenburg, Oldenburg, Germany. He has implemented several industry-sponsored projects for Samsung Electronics, Ventech Solutions, USA, Titan R D, and Oretes Consulting, respectively. He is currently working on the India-Korea bilateral project and India-Serbia bilateral, sponsored by the DST along with industry-funded research projects. He has more than 100 research papers in reputed journals and conferences at the international level. His research interests include signal processing, adaptive filters, machine learning, acoustics and audio signal analysis, and hearing-aid system designs. He was the recipient of the Marie-Curie Postdoctoral fellowship from the Department of Electronic Systems, Aalborg University, Denmark from 2015 to 2017. He is an Associate Editor for JASM and CSSP Springer.

...